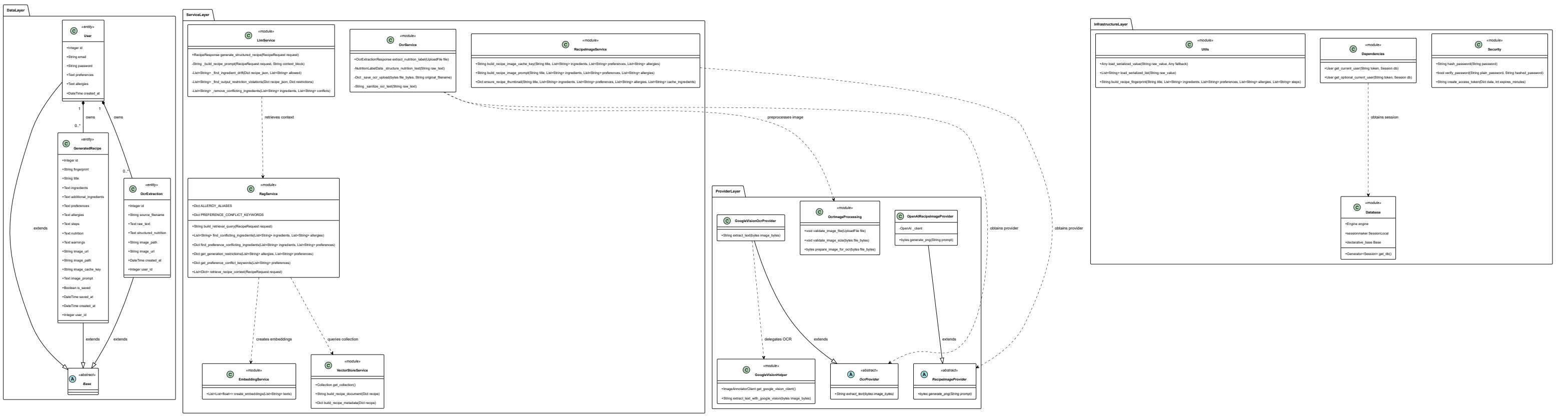
DataLayer
«entity»
User
+Integer id
+String email
+String password
+Text preferences
+Text allergies
+DateTime created_at
1
owns
0..*
1
owns
0..*
«entity»
GeneratedRecipe
+Integer id
+String fingerprint
+String title
+Text ingredients
+Text additional_ingredients
+Text preferences
+Text allergies
+Text steps
+Text nutrition
+Text warnings
+String image_url
+String image_path
+String image_cache_key
+Text image_prompt
+Boolean is_saved
+DateTime saved_at
+DateTime created_at
+Integer user_id
«entity»
OcrExtraction
+Integer id
+String source_filename
+Text raw_text
+Text structured_nutrition
+String image_path
+String image_url
+DateTime created_at
+Integer user_id
extends
extends
extends
«abstract»
Base
ServiceLayer
«module»
LlmService
+RecipeResponse generate_structured_recipe(RecipeRequest request)
-String _build_recipe_prompt(RecipeRequest request, String context_block)
-List<String> _find_ingredient_drift(Dict recipe_json, List<String> allowed)
-List<String> _find_output_restriction_violations(Dict recipe_json, Dict restrictions)
-List<String> _remove_conflicting_ingredients(List<String> ingredients, List<String> conflicts)
«module»
OcrService
+OcrExtractionResponse extract_nutrition_label(UploadFile file)
-NutritionLabelData _structure_nutrition_text(String raw_text)
-Dict _save_ocr_upload(bytes file_bytes, String original_filename)
-String _sanitize_ocr_text(String raw_text)
«module»
RecipeImageService
+String build_recipe_image_cache_key(String title, List<String> ingredients, List<String> preferences, List<String> allergies)
+String build_recipe_image_prompt(String title, List<String> ingredients, List<String> preferences, List<String> allergies)
+Dict ensure_recipe_thumbnail(String title, List<String> ingredients, List<String> preferences, List<String> allergies, List<String> cache_ingredients)
retrieves context
«module»
RagService
+Dict ALLERGY_ALIASES
+Dict PREFERENCE_CONFLICT_KEYWORDS
+String build_retrieval_query(RecipeRequest request)
+List<String> find_conflicting_ingredients(List<String> ingredients, List<String> allergies)
+Dict find_preference_conflicting_ingredients(List<String> ingredients, List<String> preferences)
+Dict get_generation_restrictions(List<String> allergies, List<String> preferences)
+Dict get_preference_conflict_keywords(List<String> preferences)
+List<Dict> retrieve_recipe_context(RecipeRequest request)
creates embeddings
queries collection
«module»
EmbeddingService
+List<List<float>> create_embeddings(List<String> texts)
«module»
VectorStoreService
+Collection get_collection()
+String build_recipe_document(Dict recipe)
+Dict build_recipe_metadata(Dict recipe)
preprocesses image
obtains provider
obtains provider
ProviderLayer
GoogleVisionOcrProvider
+String extract_text(bytes image_bytes)
«module»
OcrImageProcessing
+void validate_image_file(UploadFile file)
+void validate_image_size(bytes file_bytes)
+bytes prepare_image_for_ocr(bytes file_bytes)
OpenAIRecipeImageProvider
-OpenAI _client
+bytes generate_png(String prompt)
delegates OCR
extends
extends
«module»
GoogleVisionHelper
+ImageAnnotatorClient get_google_vision_client()
+String extract_text_with_google_vision(bytes image_bytes)
«abstract»
OcrProvider
+String extract_text(bytes image_bytes)
«abstract»
RecipeImageProvider
+bytes generate_png(String prompt)
InfrastructureLayer
«module»
Utils
+Any load_serialized_value(String raw_value, Any fallback)
+List<String> load_serialized_list(String raw_value)
+String build_recipe_fingerprint(String title, List<String> ingredients, List<String> preferences, List<String> allergies, List<String> steps)
«module»
Dependencies
+User get_current_user(String token, Session db)
+User get_optional_current_user(String token, Session db)
«module»
Security
+String hash_password(String password)
+bool verify_password(String plain_password, String hashed_password)
+String create_access_token(Dict data, int expires_minutes)
obtains session
«module»
Database
+Engine engine
+sessionmaker SessionLocal
+declarative_base Base
+Generator<Session> get_db()